Scale Drawings

Teacher's Edition

The Period

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

Must not Perimeter and area under scaling are 5.02–03 (k^2); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

Target I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

MTR **MTR.2.1** — Notes I: the same pair of figures as a picture, two side lengths and one number k · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — boards 1–3: one question three times

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	13 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A scale factor is the one number every length is multiplied by. Copy divided by original.
2. Less than 1 shrinks. More than 1 enlarges. The angles never change.
3. To the copy, multiply. Back to the original, divide.

Slide by Slide

1. **Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the lesson in miniature: a scale is a ratio, and the first move is always to read it per 1. *1 retrieves 4.06 — multiply before adding; the test was yesterday, so the first band reaches to the last lesson (yesterday). 2 retrieves 4.05 — a perfect square's root; 5.02–03 uses it to find k from two areas (last week). 3 retrieves 3.08 — reading scientific notation back (last unit). 4 retrieves MA.6.AR.3.2 — a ratio as a unit rate: 12 ÷ 3. Today's scale is exactly this, read per 1 (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. −5 2. 9 3. 2,500 4. 4 km
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the line.** Two rectangles, 6 by 4 and 9 by 6.
6. **Notes.** The direction is fixed all unit: copy ÷ original.
7. **Notes.** From the original to the copy, multiply; from the copy back to the original, divide.
8. **Example 1 — Find k from two figures.** Ask which side matches which before anyone divides: 15 matches 6, 10 matches 4.
9. **Example 1 — Worked.** 15 ÷ 6 = 2.5 and 10 ÷ 4 = 2.5: the same k from both pairs, so it is a copy and k = 2.5.
10. **Your Turn — Same steps, your numbers. Boards up when done.** k = 1.5 from both pairs (12 ÷ 8 and 15 ÷ 10).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Use a scale on a map.** Ask what the scale says per 1 inch before anyone multiplies.
13. **Example 2 — Worked.** Every inch is 25 miles, so 2.6 inches is 2.6 × 25 = 65 miles.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 63 km: 4.2 × 15.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** A 2 cm side becomes 6 cm in the copy. What is the scale factor? → ***k = 3*** *⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]*
18. **18–19. Board 2.** An 8 cm side becomes 4 cm in the copy. What is the scale factor? → ***k = ½*** *2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception]*
20. **20–21. Board 3.** A 12 cm side becomes 9 cm in the copy. What is the scale factor? → ***k = ¾*** *1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]*
22. **22–23. Board 4.** A side of 5 cm becomes 2 cm in the copy. What is the scale factor? → ***B — 0.4*** *(A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to*

get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

24–25. Board 5. The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy? → ***21 cm*** *1.7 — divided by k going to the copy [Notes III: to the copy, multiply]*

26–27. Board 6. A drawing has a 40° angle. The real figure is a scale copy at $k = 2.5$. What is the angle in the real figure? → ***A — 40°*** *(B) 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first? → ***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy*** *'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides*

30–31. Board 8. Dev, who is 13, draws a 15-ft wall as 3 in long. How long is a 20-ft wall on his drawing? → ***4 in*** *The 13 is not needed. 100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]; 8 — 20 − 15 + 3, the additive error*

32–33. Board 9. Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor. → ***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.*** *'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–3 find k, going over 1, under 1, then a fraction; 4 asks k as a vote; 5 reverses the given (k and the original, asked for the copy). The pattern breaks at 6, the one measurement a scale factor leaves alone. 7 asks whether a copy is a copy at all; 8 reads a two-unit scale per 1 unit; 9 puts the direction in writing. The independent set walks the same path in six: k over 1, k under 1, k from two pairs, a side from k, a map distance, and back to the map.

Misconceptions to Watch

Board 1. ⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]

Board 2. 2 — divided original by copy

Board 3. 1⅓ — original ÷ copy [7.GR.1.5]

Board 4. original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards

Board 5. 1.7 — divided by k going to the copy [Notes III: to the copy, multiply]

Board 6. 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]

Board 7. 'both' — checked one pair and stopped [Notes I: the same k from every side]

Board 8. 100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]

Board 9. 'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]

Materials Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.